Student's Name: .....

Arusha Regional Annual Examination December 2022

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION
FORM ONE ANNUAL EXAMINATION
BASIC MATHEMATICS

Time: 2:30 Hours **02nd DECEMBER 2022**

INSTRUCTIONS

1. This paper consists of 10 questions.
2. Attempt **ALL** questions.
3. Each question carries 10 marks.
4. Mathematical tables may be used.
5. Write your **NAME** on top of every page of your answer sheets.

1. (a) The school fee per child is 3050900 shillings per year. Juhudi paid "One million three hundred thousand and forty shillings". How much money does she owe the school?

(b) On a farm of animals, one−sixth of them are sheep, three−eighths are rabbits, a quarter is cattle, one−twelfth are goats and the rest are donkeys. If there are 152 donkeys on the farm, find the total number of animals on the farm.

MsomiBora.com

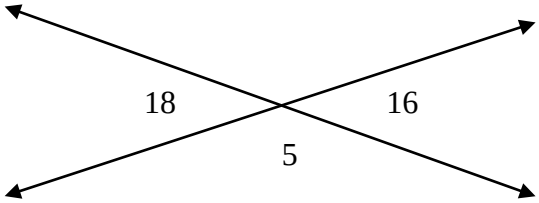
2. (a) The flights schedule at Julius Nyerere International Airport (JNIA) shows that, after every 18 minutes an airplane leaves the airport. If the fifth airplane left the airport at 5.52 p.m, at what time did the second airplane leave the airport in 24- hour clock?

(b) (i) Round off 63.9975 to 4 significant figures

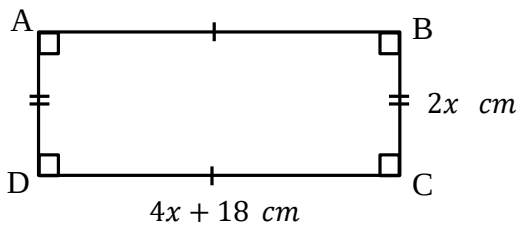
(ii) Start with 5.5 dm, copy and complete the following table with equal measures

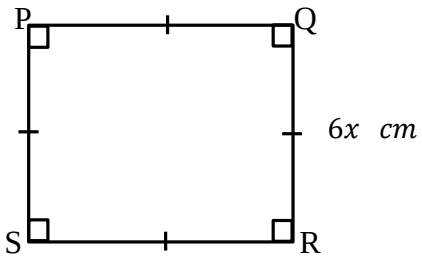
…….dam	……..cm	5.5 dm	……..m	…….km

3. (a) If m∠ 18 = y and m∠ 16 =6y-290 , find the value of m∠5 in the following figure:



(b) If the following rectangle ABCD and the square PQRS have the same perimeter, find the area of the square PQRS





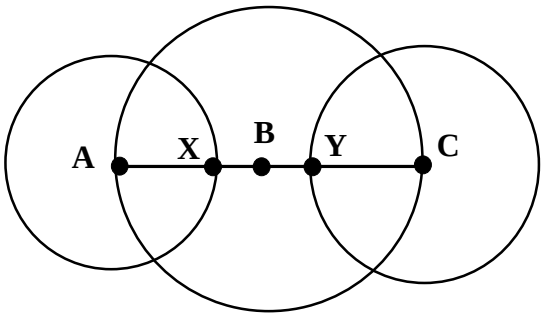
4 (a) Graph the solution of the following $|2x - 3| + 2 \leq 7$

(b) Solve the following simultaneous equations using method of substitution $\begin{cases} 6x - 4y = 24 \\ 5x - 3y = 19 \end{cases}$

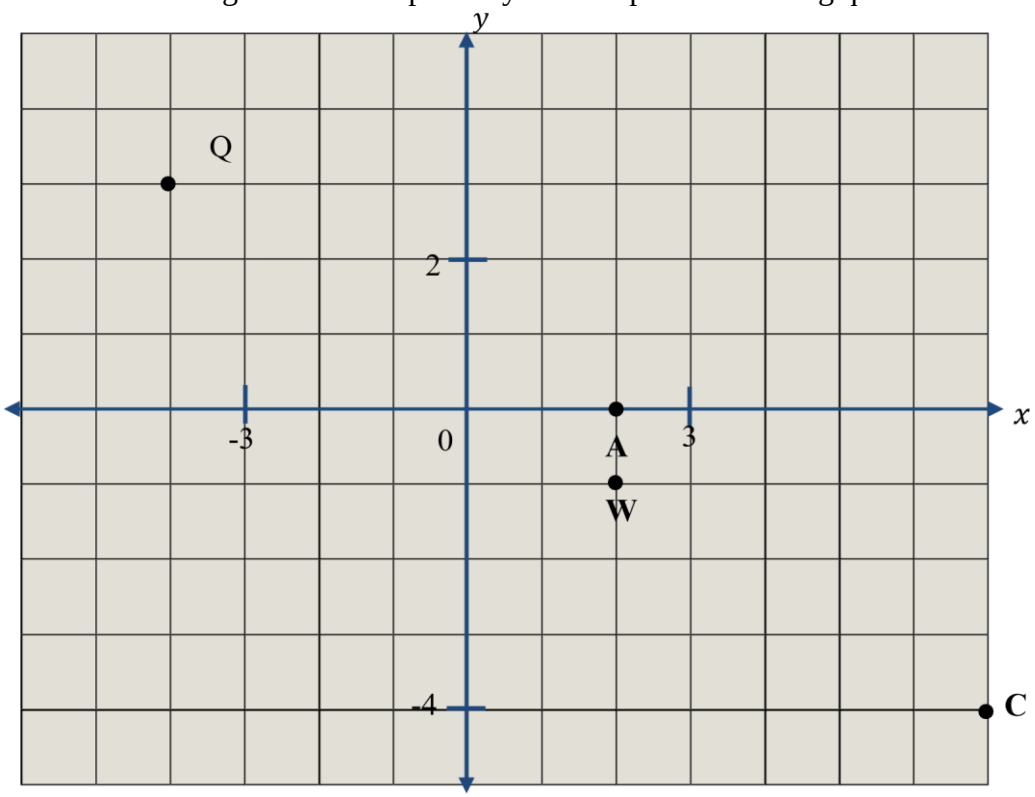
5. (a) Seypalm bought a second hand car at 4 400 000 shillings and incurred some cost for its repairing. After a while he sold it at 5 600 000 shillings and obtained a profit of 12%. Find how much money he incurred for its repair.

(b) Share 8 million into the ratio of 2: 5:3. Leave the answer into whole numbers

6. (a) The diametres of circle A, circle B, and circle C are 10 cm, 20 cm, and 14 cm respectively. Calculate length XB and length BY



(b) Use the following Coordinates plane xy to attempt the following questions:



(i) Write the quadrant and the coordinate of point P

(ii) Find the "$y = mx + c$ "of the line PC if point P is consecutive point with point A on the y-axis.

7. (a) (i) Write the place value of a prime digit in the following numeral 90 284.

(ii) Convert $2.3\dot{5}$ as a ratio of two integers where the denominator is not equal to zero.

(b) Sarah got 85% in mathematics test that had 40 questions. How many questions did she get correctly?

8. (a) Find the area of three-quarters a circle whose circumference is 47 cm. (use $\pi = 22/7$)

(b) Find the sum of all prime numbers between 120 -130.

9. (a) Solve graphically the following simultaneous equations

$$2x - y = 10$$
$$x + y = -1$$

(b) The sum of two numbers is 11 while their difference is 5. Find the numbers

10. (a) (i) How many real numbers are there between 5 -7?

(ii) In a class of 40 students, 34 students are present. Write the first two equivalent fractions of the students who are absent where the numerator is less than 10.

(iii)A shop keeper sold 192 shirts at the price of 6970 shilling each. Estimate how much money he got.

(b) Change 5.7 litres into cubic centmetres